

High-frequency data and limit order books



Lecture 2 - February 3rd, 2026

Ioane Muni Toke

Université Paris-Saclay
CentraleSupélec, MICS Laboratory, FiQuant Team



CentraleSupélec

U. Paris-Saclay CentraleSupélec cursus Ingénieur 3A Mathématiques et Data Science
IP Paris ENSAE M2 Statistics, Finance and Actuarial Science
U. Paris-Saclay Evry M2 Quantative Finance

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General information on labs

Lab 1

Labs - General information

- ▶ You should work by **groups of three** students.
- ▶ The expected output of each lab is **a single Python notebook** with code, graphs and comments/analyses **Notebooks must be fully executable** from top to bottom in one run.
- ▶ For the lab i , your notebook **MUST** be named **DHF-LAB i -NAME1-NAME2-NAME3.ipynb**, where NAME1 and NAME2 and NAME3 are the names of the students of the group in capital letters (and without space).
- ▶ For each lab, a dedicated space will be opened on the **Edunao** page of the course. **Upload your notebook only**, nothing else (no data, no saved graphs, no pdf report).
- ▶ If all students in the group are non-CS students (**and only in this case**), an email submission received by the deadline will be accepted.
- ▶ Each lab will have a specific deadline, please check it.

Labs - About the data provided

- ▶ Data used in the labs of this course is not public. This is research data provided via the FiQuant research team (MICS, CentraleSupélec, Université Paris-Saclay).
- ▶ The use of such data is restricted to teaching and research purposes.
- ▶ Data will be available on Edunao until the end of the course. You must not keep any copy of this data after the course has ended.
- ▶ **You must not use this data for any purpose other than the labs of this course. Keep a local copy only. You must not share this data. This data must not be uploaded to any github, gitlab, or any other platform.**

Labs - Dos... I

- ▶ **Keep the file structure of the data.** Data files `stocki_dayj_xxx.csv.gz` must be placed in a directory `Data` at the root of your Jupyter server, with your notebook.
- ▶ Decompress the global archive, but **do not decompress individual data files.** `pandas` can directly read `.csv.gz` files.
- ▶ **Convert timestamps to datetime objects** with `pd.to_datetime(...)`. `Matplotlib` deals well with `datetime` and this is necessary for efficient sampling of the data when needed.
- ▶ Depending on your command of `Python/pandas` and the computing/memory resources of your computer, you may decide to develop/debug **on a small subsample of the available data** (e.g. a few days first), then use the full sample for final graphs and analyses.

Labs - Dos... II

- ▶ Labs are made of open questions. These questions are guides, but you can go further. For each question, you are expected to perform some data analysis or numerical experiment and show the results using plots.
- ▶ All graphs should have a legend, a title, labeled x-axis and y-axis with units if applicable.
- ▶ All graphs should be explained (how did we compute it ?), analyzed (what do we see) and commented (expected ? literature ?).
- ▶ Comments can be written in either French or English **in markdown cells only**.
- ▶ If you use bibliographic references in your comments, please cite them fully and correctly.

Labs - ...and don'ts

- ▶ **Don't** leave any system command (pip install ...) in your notebooks. They will be blocked, your notebook will not run and your lab will not be graded.
- ▶ **Don't** use plotly (html/pdf conversion problems). Please use matplotlib instead.
- ▶ **Don't** use horizontal scrolling in your code cells. Please break your lines of code instead (in your IDE, add a vertical ruler at 80 characters).
- ▶ **Don't** plot log of (physical) quantities. Please use log scales instead.
- ▶ **Don't** leave plot randomly in the notebook without context. Please indicate question numbers and comment on your results **in markdown cells** (and if your results are not what you were expecting, say it !).
- ▶ **Don't** leave output printed for development/debugging purposes, but useless for the final version (in particular any progress counter, progress bar, etc.).

Labs - Grading process

- ▶ The grading process of the labs is the following:
 1. Notebooks are converted to python scripts
 2. Python scripts are parsed by a plagiarism detection software.
 3. Notebooks are cleared of all outputs, fully reexecuted with clean kernels, then exported to html, and finally converted to pdf. Notebooks with execution failures will not be graded.
- ▶ Because of this grading pipeline:
 - ▶ Avoid complex latex formulas in your markdown comments.
 - ▶ Make sure your notebooks are fully executable: clear all outputs, kill and restart the kernel and finally run all cells in one pass before submitting.
 - ▶ Make sure your graphs are correctly exported to html.
- ▶ You're expected to work in **Python with standard scientific and data libraries**. Notebooks will be executed in a (conda) Python environment containing only python 3.12.2 and its standard library, jupyter, numpy, scipy, pandas, polars, matplotlib, powerlaw, statsmodels, Hawkes.

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General information on labs

Lab 1

Lab 1 - Questions I

1. **Trades.** This part deals with trades data only.
 - 1.1 **Dataset.** Perform general statistical analyses on the datasets. Are the results expected ? Is the dataset free of visible problems ?
 - 1.2 **Tick sizes.** Measure the tick sizes on your dataset. Comment.
 - 1.3 **Intraday activity.** Measure the intraday trading activity on your dataset. Comment.
 - 1.4 **Volatility and traded volume.** Is there a link between volatility and traded volume in your dataset ?
 - 1.5 **High-frequency log-returns.** Compute the distributions of log-returns in your dataset for various sampling frequencies in calendar time (each observed value is weighted by the length of the time interval on which it is observed) and event time (each observed value counts for one observation). Comment.

Lab 1 - Questions II

- 2. **Quotes.** This part deals with the updates data only.
 - 2.1 **Dataset.** Perform general statistical analyses on the datasets. Are the results expected ? Is the dataset free of visible problems ?
 - 2.2 **Prices.** Plot the best bid price, best ask price, mid price and weighted mid-price on a 5-minute window of your sample.
 - 2.3 **Spread.** Compute the empirical distribution of the spread. Comment. Compare the distribution in event time and in calendar time. Comment. Would you say that, on your data, the spread is stationary throughout the trading day ?
 - 2.4 **Imbalance and mid-price moves.** Would you say that, on your data, the imbalance is a good predictor of the next (non-zero) mid-price move ? What about the spread ?

Lab 1 - Questions III

3. **Trades and quotes.** (if you have time, to take it further).

- 3.1 **Dataset.** Are the trades and updates files coherent ? Can you match the trades of the trades files with the update data ? Sign the trades.
- 3.2 **Imbalance and trades.** Would you say that, on your data, the imbalance is a good predictor of the next trade sign ? What about the spread ?
- 3.3 **Autocorrelations.** Compute the autocorrelation function of the trade signs. Comment.